

Prof. Dr.-Ing. Viktor Wesselak

Research

Lecture Scientific Practice

Overview

1. Research on the Internet

Search engines, search strategies, blind spots, relational databases, special case Wikipedia

2. Search for books

Libraries and Databases for Engineering Sciences - Search Techniques - Content indexing of a library - Keywords

3. Search for journal articles

Journals and databases for the engineering sciences - Content indexing of a journal database - Handling of thesauri

4. Further researches

Search for standards, patents and suppliers

1. Research on the internet - search engines

Search engines help to make information available on the Internet, which currently (2018) consists of an estimated 2 billion websites.

Search engines: www.google.de (85% worldwide market share of pageviews)

www.bing.de (5%, owned by Microsoft)

www.yahoo.de (3%, owned by Verizon)

www.web.de (uses google)

www.baidu.com (1% worldwide, almost 60% in China)

How search engines work:

➡ Catalog (former)

➡ Robots + portal for mail services, news, shopping

1. Research on the Internet - Google

The single line editor:
Enter a question and
get a result!



➡ But which one?
➡ And why?

Google-Suche

Auf gut Glück!

Example: Search for "solar panel" and "solar panels"
Search for "photovoltaics" and "photovoltaic"

➡ And here I stand, with all my lore,
Poor fool, no wiser than before. (Goethe)

1. Research on the Internet - Google search queries

- The AND operation: more terms result in (theoretically) fewer hits - the google standard for search queries consisting of several terms
explicitly: AND or &
- The OR operation: more terms result in (theoretically) more hits
explicit: OR or |
- Formation of phrases: the search is for an exact sequence of terms (theoretically):
explicit: "Term1 Term2 Term3"

- ➡ Search query 1: (Victor | Viktor) & Wesselak
All pages are searched on which the term pairs "Viktor" and "Wesselak" or "Victor" and "Wesselak" can be found.
- ➡ Search query 2: "Viktor Wesselak" | "Victor Wesselak"
All pages are searched on which the terms "Viktor Wesselak" or "Victor Wesselak" can be found.

1. Research on the Internet - Google sorting

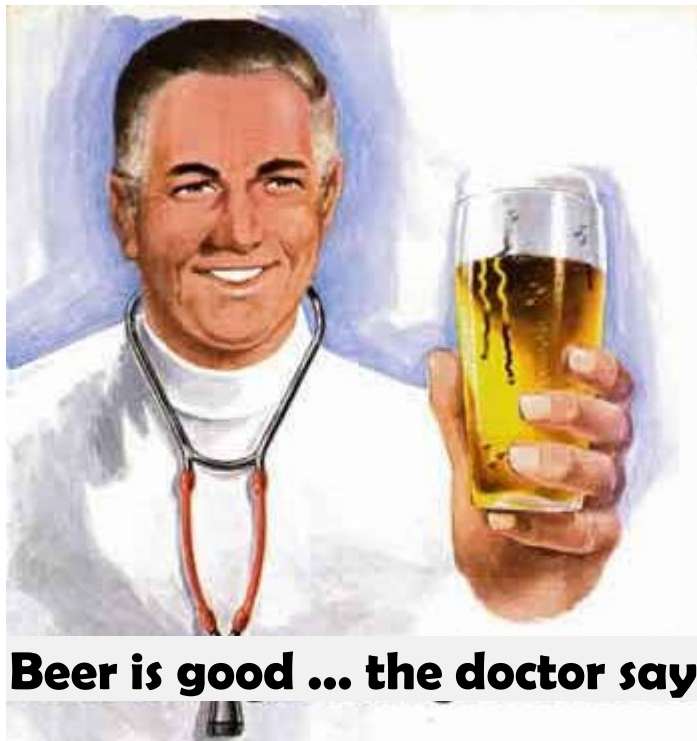
*"There is nothing better to hide
than on the second Google hit page,,*

(old Klingon proverb)

Obviously it makes a difference whether a search result is displayed first or thirty-four thousand one hundred and seventy-two. So: how is the ranking of the search results achieved?

Room for conspiracy theories:

1. Research on the internet - blind spots



Source: www.bier-fibel.de/bier-gegen-entzundungen/

Beer is good ... the doctor says!



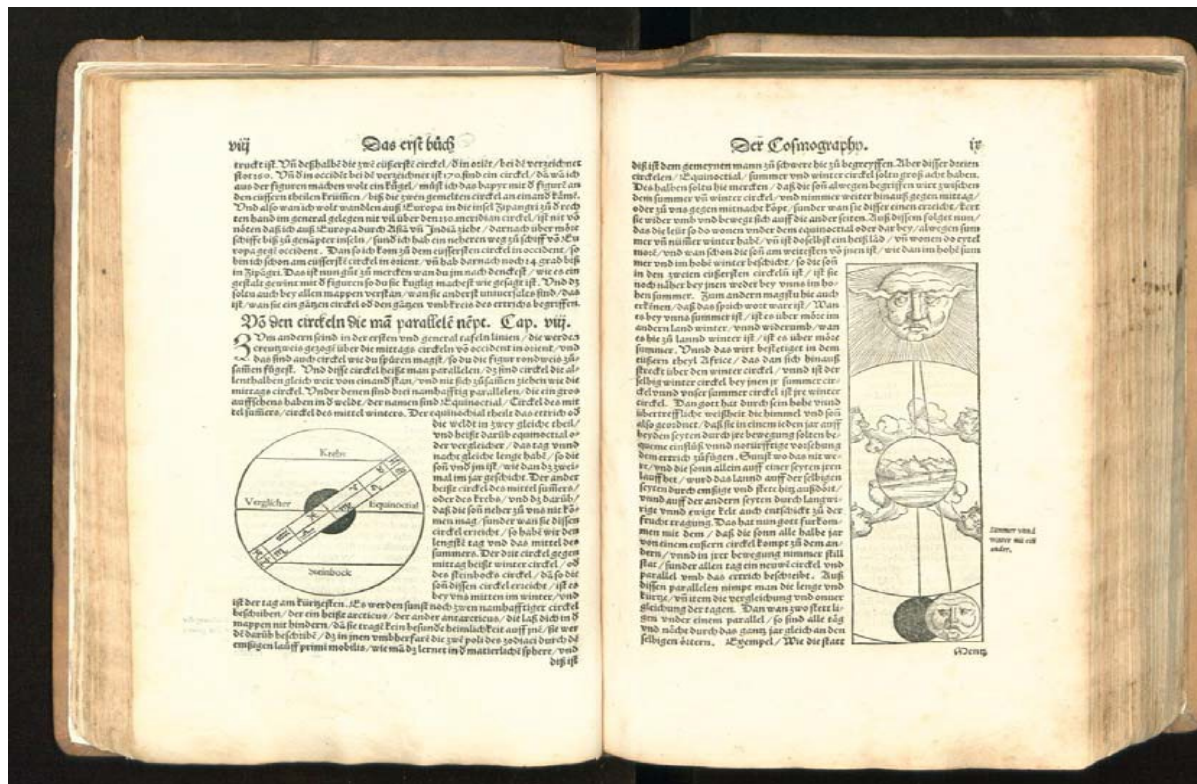
Source: www.mz-web.de/weissenfels/

Statements cannot be separated from their sources!



You need a critical assessment of sources

1. Research on the internet - blind spots



Source: Dusseldorf University Library

The Internet exists only since yesterday - it is often blind for Information from the time before!

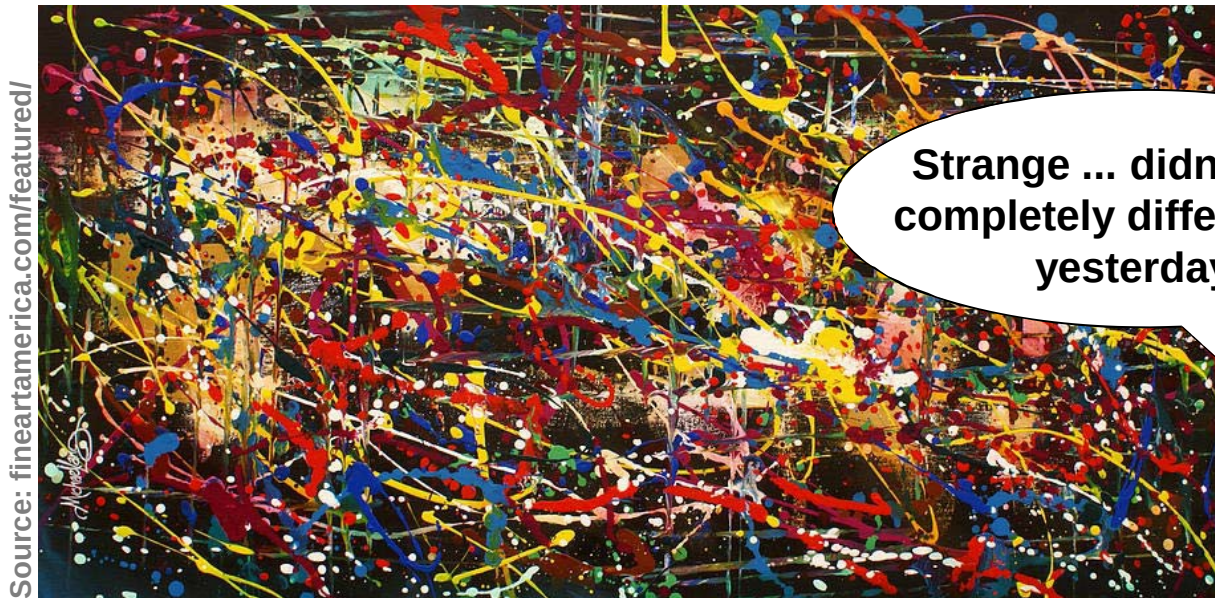
1. Research on the internet - blind spots



Source: Getty Images/Montage "The World"

Google (Bing/Yahoo/Web.de) is not committed to the truth,
but the profit expectations of its shareholders!

1. Research on the internet - blind spots



Strange ... didn't it look completely different here yesterday?

The Internet is not a "hard" scientific source, because it changes permanently. Sooner or later every link to an internet source will be a "dead link".



1. Research on the internet - blind spots



Workaround 1:



www.archive.org/web/

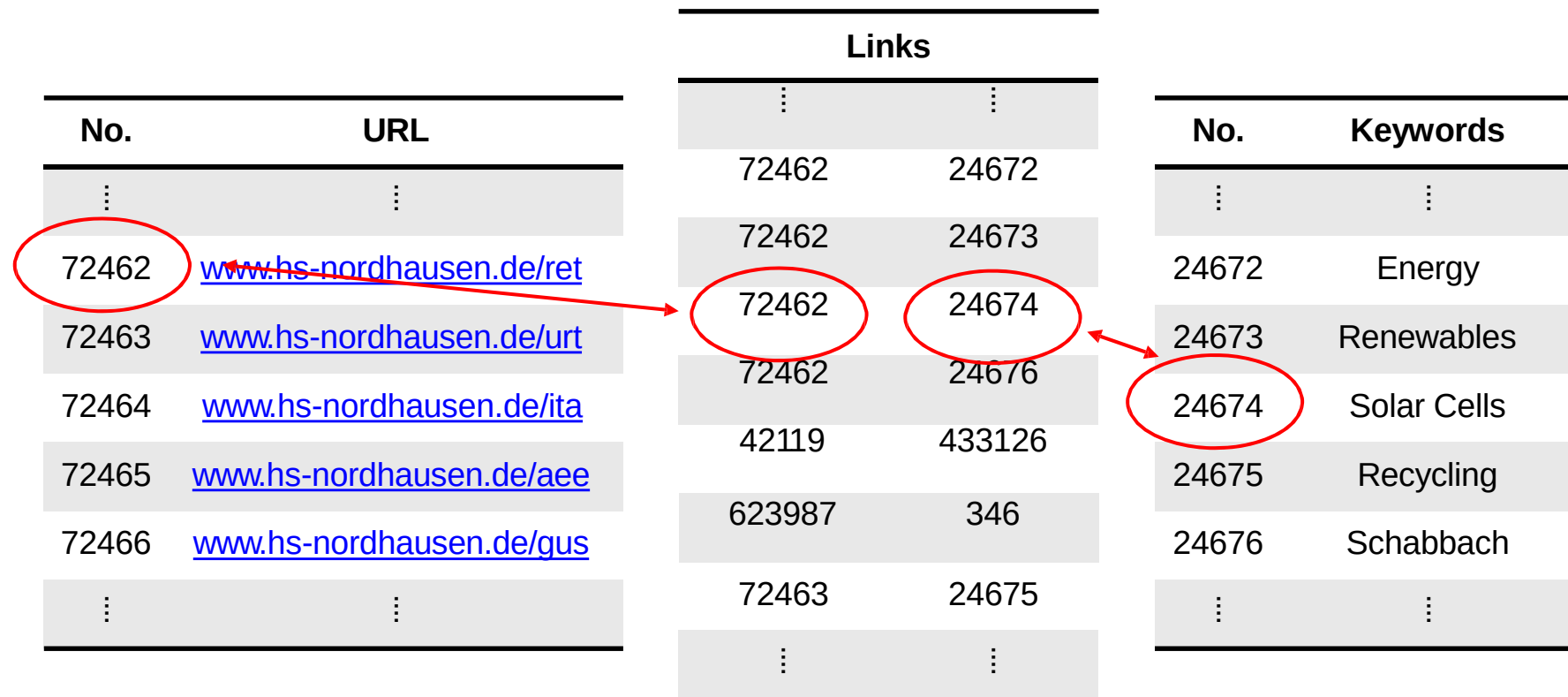
Workaround 2: Cite DOI (Digital Object Identifier) sources



Shortened: The DOI refers to a database in which the current URL of the corresponding object can be found. DOIs are mainly used by scientific publishers for digital publications.

1. Research on the Internet - Relational databases

Information is stored in databases. An efficient variant are so-called relational databases:



1. Research on the Internet - Relational databases

Google stores its information in multidimensional tables, which can contain a timestamp, for example.

No.	Timestamp	URL
⋮	⋮	
72462.10	2018.03.12	
72463.32	2018.04.14	
72464.10	2018.04.22	
72465.29	2017.12.30	
⋮	⋮	

No.	Timestamp	URL
⋮	⋮	
72462.11	2018.04.22	
72463.33	2018.04.21	
72464.11	2018.04.25	
72465.30	2018.04.20	
72466.1	2018.03.23	
⋮	⋮	

No.	Timestamp	URL
⋮	⋮	⋮
72462.12	2018.04.24	www.hs-nordhausen.de/ret
72463.34	2018.04.23	www.hs-nordhausen.de/urt
72464.12	2018.04.27	www.hs-nordhausen.de/ita
72465.31	2018.04.23	www.hs-nordhausen.de/aee
72466.2	2018.03.31	www.hs-nordhausen.de/gus
⋮	⋮	⋮

1. Research on the Internet - Wikipedia

Wikipedia appears since 2001 in the Internet and is free and advertising-free. It is financed by donations. There are over 2.000.000 keywords in the German Wikipedia alone.

Approximately 6.700 registered but unknown regular authors in the German Wikipedia.

Wikipedia killed all commercial printed encyclopedia projects (e.g. Encyclopedia Britannica).

problems:

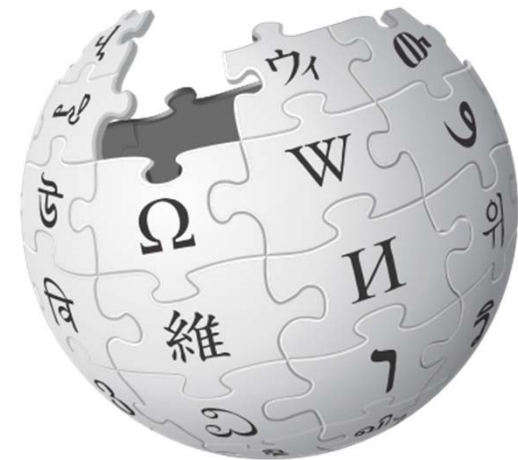


no static page content but a traceability of changes



Edit-wars (Example: [Saxony](#))

Wikipedia is no scientific source!



WIKIPEDIA
Die freie Enzyklopädie

2. Search for books - the HSN Library

The library of the HS Nordhausen has an up-to-date and well-equipped range of books for the study programs offered here.

Access is via the OPAC (Online Public Access Catalogue) - behind it, of course, is a database.



The screenshot shows the OPAC (Online Public Access Catalogue) interface of Hochschule Nordhausen. At the top, there is a navigation bar with links: Home, Weitere Kataloge, Suchen (highlighted in red), Suchergebnis, Erweiterte Suche, Zwischenablage, Benutzerkonto, and Hilfe. Below this, there is a search area with a dropdown menu for 'suchen [und]', a text input field containing '[ALL] Alle Wörter', a dropdown for 'sortiert nach' set to 'Erscheinungsjahr', and a checkbox for 'unscharfe Suche'. A 'Suchen' button is located to the right of the input field. On the left side, there is a vertical sidebar with the Hochschule Nordhausen logo and a link labeled 'Abmelden'. The main content area features the title 'OPAC der Hochschule Nordhausen' in bold, followed by a paragraph: 'Im OPAC werden alle Bücher, Zeitschriften sowie die elektronischen Medien der Bibliothek der Hochschule Nordhausen nachgewiesen. Als angemeldeter Bibliotheksbenutzer haben Sie die Möglichkeit, über den Menüpunkt "Benutzerkonto" ausgeliehene Medien zu verlängern.' Below this, there is a note: 'Bitte beenden Sie Ihre Arbeit im OPAC über den Menüpunkt "Abmelden".' and a reminder: 'Erinnerung per E-mail vor Ablauf der Leihfrist: [Bücherwecker](#)'.

2. Search for books - the HSN Library

■ Ihre Aktion suchen [oder] ([ALL] Alle Wörter) solarzelle

	Titel: Photovoltaik wie Sonne zu Strom wird / Viktor Wesselak , Sebastian Voswinckel
Person/en:	Wesselak, Viktor, 1965- [VerfasserIn] Voswinckel, Sebastian [VerfasserIn] Springer-Verlag GmbH [Verlag]
Körperschaft/en:	
Ausgabe:	2. Auflage
Sprache/n:	Deutsch
Veröffentlichungsangabe:	Berlin ; Heidelberg : Springer, [2016]
Umfang:	X, 135 Seiten : Illustrationen, Diagramme
Schriftenreihe:	Technik im Fokus
ISBN:	3-662-48905-8, 978-3-662-48905-5 Weitere Ausgaben: 978-3-642-24296-0
Schlagwörter:	* Fotovoltaik / Solarzelle / Fotovoltaik
Sachgebiete:	52.56 Regenerative Energieformen al 53.36 Energiedirektumwandler elektr
Mehr zum Thema:	Dewey Dezimal-Klassifikation: [23sdr] Regensburger Verbund-Klassifikation
Mehr zum Titel:	http://www.springer.com/ [Verlag] Inhaltsverzeichnis Inhaltsbeschreibung
Sachgebiete:	ZP 3730
Standort:	Freihandbestand
Signatur:	ZP 3730 W515(2)
Ausleihstatus:	Nicht ausleihbar Bitte selbst am Standort Ebene 1 entnehmen.

How do I find what I am looking for?

Search for keywords:

Words in title (Care: negative lists)

Names

Publisher name

Series title

2. Search for books - the HSN Library

■ Ihre Aktion suchen [oder] ([ALL] Alle Wörter) solarzelle

	Titel:	Photovoltaik : wie Sonne zu Strom wird / Viktor Wesselak, Sebastian Voswinckel
	Person/en:	Wesselak, Viktor, 1965- [VerfasserIn] Voswinckel, Sebastian [VerfasserIn]
	Körperschaft/en:	Springer-Verlag GmbH [Verlag]
	Ausgabe:	2. Auflage
	Sprache/n:	Deutsch
	Veröffentlichungsangabe:	Berlin ; Heidelberg : Springer, [2016]
	Umfang:	X, 135 Seiten : Illustrationen, Diagramme
	Schriftenreihe:	Technik im Fokus
	ISBN:	3-662-48905-8, 978-3-662-48905-5 Weitere Ausgaben: 978-3-642-24296-*
	Schlagwörter:	*Fotovoltaik / Solarzelle / Fotovoltaik
	Sachgebiete:	52.56 Regenerative Energieformen al 53.36 Energiedirektumwandler elektri
	Mehr zum Thema:	Dewey Dezimal-Klassifikation: [23sdr Regensburger Verbund-Klassifikation:
	Mehr zum Titel:	http://www.springer.com/ [Verlag] Inhaltsverzeichnis Inhaltsbeschreibung
	Sachgebiete:	ZP 3730
	Standort:	Freihandbestand
	Signatur:	ZP 3730 W515(2)
	Ausleihstatus:	Nicht ausleihbar Bitte selbst am Standort Ebene 1 entnehmen.

How do I find what I am looking for?

Search for keywords:

With keywords a contentwise evaluation of database entries is performed. It is worth getting into the perspective of the keywriter.

2. Search for books - the HSN Library

■ Ihre Aktion suchen [oder] ([ALL] Alle Wörter) solarzelle

	Titel:	Photovoltaik : wie Sonne zu Strom wird / Viktor Wesselak, Sebastian Voswinckel
	Person/en:	Wesselak, Viktor, 1965- [VerfasserIn] Voswinckel, Sebastian [VerfasserIn]
	Körperschaft/en:	Springer-Verlag GmbH [Verlag]
	Ausgabe:	2. Auflage
	Sprache/n:	Deutsch
	Veröffentlichungsangabe:	Berlin ; Heidelberg : Springer, [2016]
	Umfang:	X, 135 Seiten : Illustrationen, Diagramme
	Schriftenreihe:	Technik im Fokus
	ISBN:	3-662-48905-8, 978-3-662-48905-5 Weitere Ausgaben: 978-3-642-24296-6
	Schlagwörter:	* Fotovoltaik / Solarzelle / Fotovoltaik
	Sachgebiete:	52.56 Regenerative Energieformen al 53.36 Energiedirektumwandler elektr
	Mehr zum Thema:	Dewey Dezimal-Klassifikation: [23sdr Regensburger Verbund-Klassifikation: ZN 8190 ; ZP 3730
	Mehr zum Titel:	http://www.springer.com/ [Verlag] Inhaltsverzeichnis Inhaltsbeschreibung
	Sachgebiete:	ZP 3730
	Standort:	Freihandbestand
	Signatur:	ZP 3730 W515(2)
	Ausleihstatus:	Nicht ausleihbar Bitte selbst am Standort Ebene 1 entnehmen.

How do I find what I am looking for?

The [Regensburg Classification](#) provides a fixed framework for the classification of scientific literature.

2. Search for books - the TIB Hannover

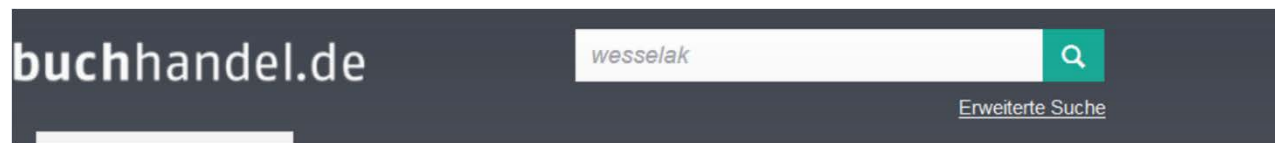
Academic libraries are often specialized in one collection area. For the natural and engineering sciences this is the TIB Hannover. There you will find:

- ➡ All German scientific and engineering dissertations
- ➡ (Almost) all natural and engineering technical journals
- ➡ Research reports from publicly funded projects
- ➡ and of course books, books, books

2. Search for books - the book trade

The German-language available books can be searched in the public search tool of the VLB database

VLB = List of available books



2. Search for books - the book trade

Books that are no longer available are traded through antiquarian bookshops. There are several databases where antiquarian bookshops have placed their stocks:

- ➡ zvab.com (mainly German language literature)
- ➡ abebooks.com (mainly English literature)
- ➡ ilab.org (meta search engine)

3. Search for journal articles

Journal articles are also indexed in common databases. A distinction is made between offers from scientific publishers, e.g.

- ➡ ScienceDirect (Elsevier publishing house)
- ➡ SpringerLink (Springer view path)

and cross-publishing, topic-oriented databases such as

- ➡ WTi Frankfurt databases
- ➡ TIB Hannover search engine

These are commercial offers.

3. Search for journal articles - WTi

The WTi database contains various sub-databases containing references to journal articles, conference proceedings and research reports.

These contain the abstracts and a content indexing by means of a comprehensive keyword catalog (thesaurus).

The WTi database is available on the HSN campus free of charge.



3. Search for journal articles - WTi

A thesaurus is a hierarchical technical vocabulary which describes a scientific field as completely as possible.

Keywording is done using the thesaurus, so searches should also be done using the thesaurus!

Example: the thesaurus of the WTi database



<http://de.grunlo.wikia.com>

Tyranno-Thesaurus-Rex

4. Other databases

Other databases for specific technical information are

- ➡ www.dpma.de (German Patent and Trademark Office)
- ➡ www.perinorm.com (Technical Standards)
- ➡ www.wlw.de (industrial products)